

MedX AI — Project Report

A chest X-ray disease prediction, explainability, and reporting platform.

1. Executive Summary

MedX AI is an end-to-end medical imaging pipeline: a clinician uploads a chest X-ray, a ResNet-50 model predicts probabilities across 14 thoracic conditions, Grad-CAM produces a visual explanation of where the model looked, an LLM turns the raw probabilities into a narrative report, and the result is persisted and retrievable per patient — all behind a REST API with a working Streamlit frontend, containerized with Docker Compose, and continuously tested via GitHub Actions.

The project is tagged `v1.1.0` on `main`. Every stage of the pipeline is real and wired, not scaffolding: the model has been trained on genuine chest X-rays (not just synthetic data), the API has been exercised end-to-end against a real Postgres database in real Docker containers, and the Grad-CAM output has been visually inspected, not just unit-tested. It was built as a sequence of small, reviewed pull requests — infrastructure and data model before any AI code — which is documented in full in `roadmap.md` (`roadmap.md`).

What this report will not claim: that the model is clinically accurate, or that this is a finished medical product. Section 13 and `known-limitations.md` (`known-limitations.md`) say plainly what isn't done yet.

2. Problem Statement

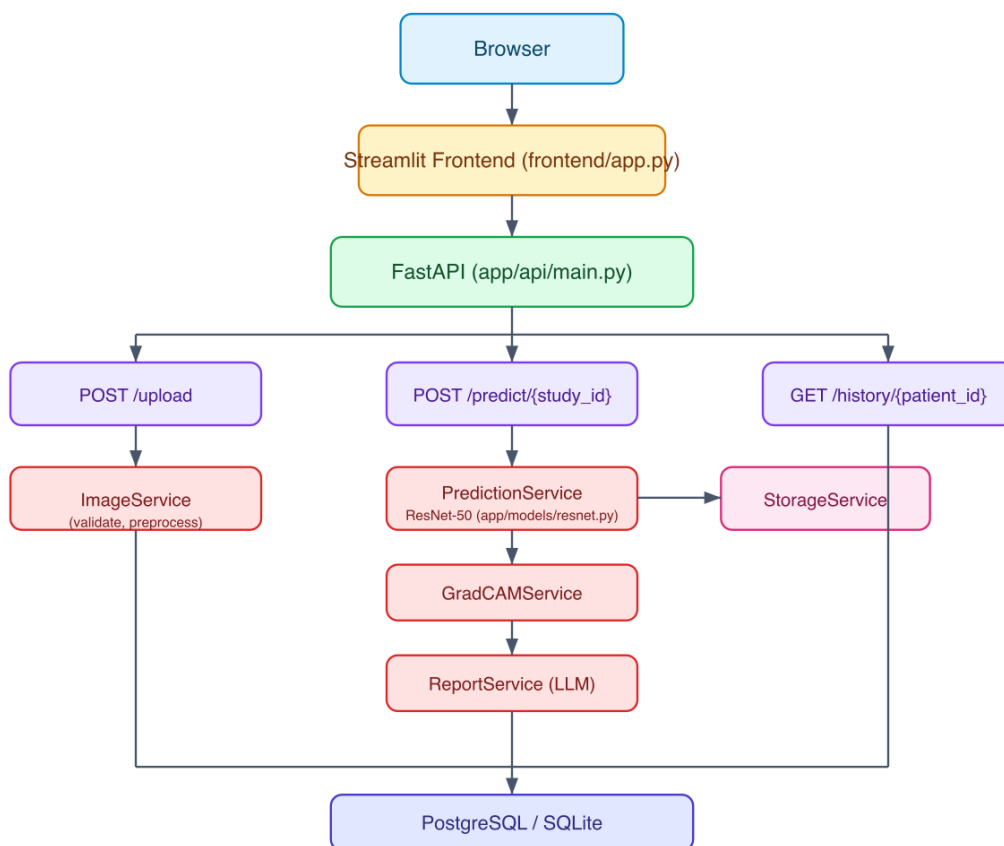
Chest X-rays are one of the most common diagnostic imaging studies, and manual review is slow relative to imaging volume. Automated triage/classification tools can help, but two things routinely go missing from prototype AI diagnostic tools: **explainability** (a probability score with no indication of *why* is not something a clinician can act on) and **a usable delivery mechanism** (a model in a notebook is not a tool anyone can use). MedX AI addresses both: every prediction ships with a Grad-CAM localization and a narrative report, delivered through a real API and UI, not a script that has to be re-run by hand.

3. Objectives

- Predict the presence of 14 thoracic conditions from a chest X-ray image (multi-label, not single-diagnosis).
- Explain each prediction visually (Grad-CAM), not just numerically.
- Generate a clinician-readable narrative report from the raw prediction, not just a probability table.

- Expose all of the above through a versioned REST API with real request/response contracts (documented in `api.md` (`api.md`)), not an ad hoc script interface.
- Persist every prediction so a patient's history can be retrieved later.
- Containerize the whole system so it runs the same way in development and in a production-shaped environment (Postgres, multi-worker serving, CI).
- Be honest, in the documentation, about what has and hasn't actually been validated.

4. System Architecture



User -> Streamlit Frontend -> FastAPI -> Inference Pipeline -> ResNet-50 -> Grad-CAM -> LLM Report -> Postgres/SQLite -> API Response -> Frontend

The system is layered so that no layer knows about the internals of the one two steps away — the API routes never touch `torch` or `PIL` directly, and the model/Grad-CAM/report services never touch HTTP. Full component and sequence diagrams, and the reasoning behind the layering, are in `architecture.md` (`architecture.md`) and `system-design.md` (`system-design.md`); the exact call sequence for a prediction is in `sequence-predict.svg` (`images/sequence-predict.svg`).

5. Technology Stack (and why)

Technology	Role	Why (full reasoning in <code>system-design.md</code> (<code>system-design.md</code>))
FastAPI	REST API	Async, Pydantic-validated request/response contracts, free auto-generated OpenAPI docs — chosen over Flask for exactly the parts of this project that need typed contracts and multiple routers.
ResNet-50 (torchvision, ImageNet-pretrained)	Disease prediction	Transfer learning makes a small dataset viable at all; an established, published baseline for this exact task (multi-label chest X-ray classification) rather than an untested architecture choice.
Grad-CAM (<code>pytorch_grad_cam</code>)	Explainability	Post-hoc — explains the trained model without changing its architecture or retraining it; the standard method in published medical-imaging XAI work.
OpenAI API (configurable)	LLM report generation	Turns 14 raw probabilities into the narrative format a radiology report actually uses; designed to fail gracefully (<code>report_text: null</code>), never to block a diagnosis on an LLM being reachable.
PostgreSQL (prod) / SQLite (dev+test default)	Persistence	Postgres for real concurrent-write correctness in production; SQLite for zero-setup local development and fast, isolated tests — identical schema either way via SQLAlchemy + Alembic.
Alembic	Schema migrations	Reviewable, reversible schema changes instead of implicit <code>create_all()</code> calls that can silently diverge between environments.
Streamlit	Frontend	Fastest path to a usable UI on top of a Python ML stack, without a separate frontend build/deploy pipeline.
Docker / Docker Compose	Deployment	One <code>docker-compose up</code> brings up API + Postgres + frontend together, matching production shape locally.
pytest + GitHub Actions	Testing / CI	Every push and PR to <code>main</code> / <code>develop</code> runs the full suite automatically — see Section 11.

6. Dataset

Training uses **NIH ChestX-ray14** — 14 thoracic conditions, the same taxonomy used throughout this project's `config.yaml` and database schema. Because Kaggle credentials and the official NIH box.com host weren't accessible in the build environment, `scripts/download_sample_data.py` pulls a real subset (~2,000 images with genuine multi-label annotations) from a public, no-credential-required Hugging Face mirror instead — see `model.md` (`model.md`) for the exact

approach and roadmap.md (roadmap.md)'s `feature/real-data` entry for how that dead end was found and worked around. `DummyChestXrayDataset` (random tensors) exists separately for fast pipeline smoke-testing without any network dependency — it is never used to make a claim about model accuracy.

Honest framing: 2,000 images is a meaningful pipeline-validation sample, still not a training set sized for a clinically usable model. See Section 12 and 13.

7. Model Development

Dataset -> Preprocessing -> Training -> Validation -> Checkpoint -> Inference

ResNet-50 (ImageNet-pretrained) with its final layer replaced for 14-way multi-label output (`BCEWithLogitsLoss` with per-class `pos_weight` for class-imbalance correction, independent sigmoid per class — not softmax, since a single X-ray can show multiple findings at once). Adam optimizer with a `CosineAnnealingLR` schedule. Preprocessing: RGB conversion, resize to 224×224, ImageNet normalization, with random-crop/flip augmentation during training only.

Validated with per-label and averaged AUC (`sklearn.roc_auc_score`) on an 80/20 held-out split. A real, seeded training run against 2,000 real images for 10 epochs produced **best val AUC 0.71 (epoch 4)** — genuine learning, on genuine data, far short of the documented target ($\text{AUC} \geq 0.90$, matching published full-dataset baselines). The model reproducibly overfits after ~3-5 epochs at this scale — seen on four separate runs, which is why the checkpoint saved to disk is now the *best* epoch by val AUC, not simply the last one. Full detail, including a real bug a code review caught (training augmentation was silently never applied, from the very first preprocessing implementation), in model.md (model.md).

8. Explainability (Grad-CAM)

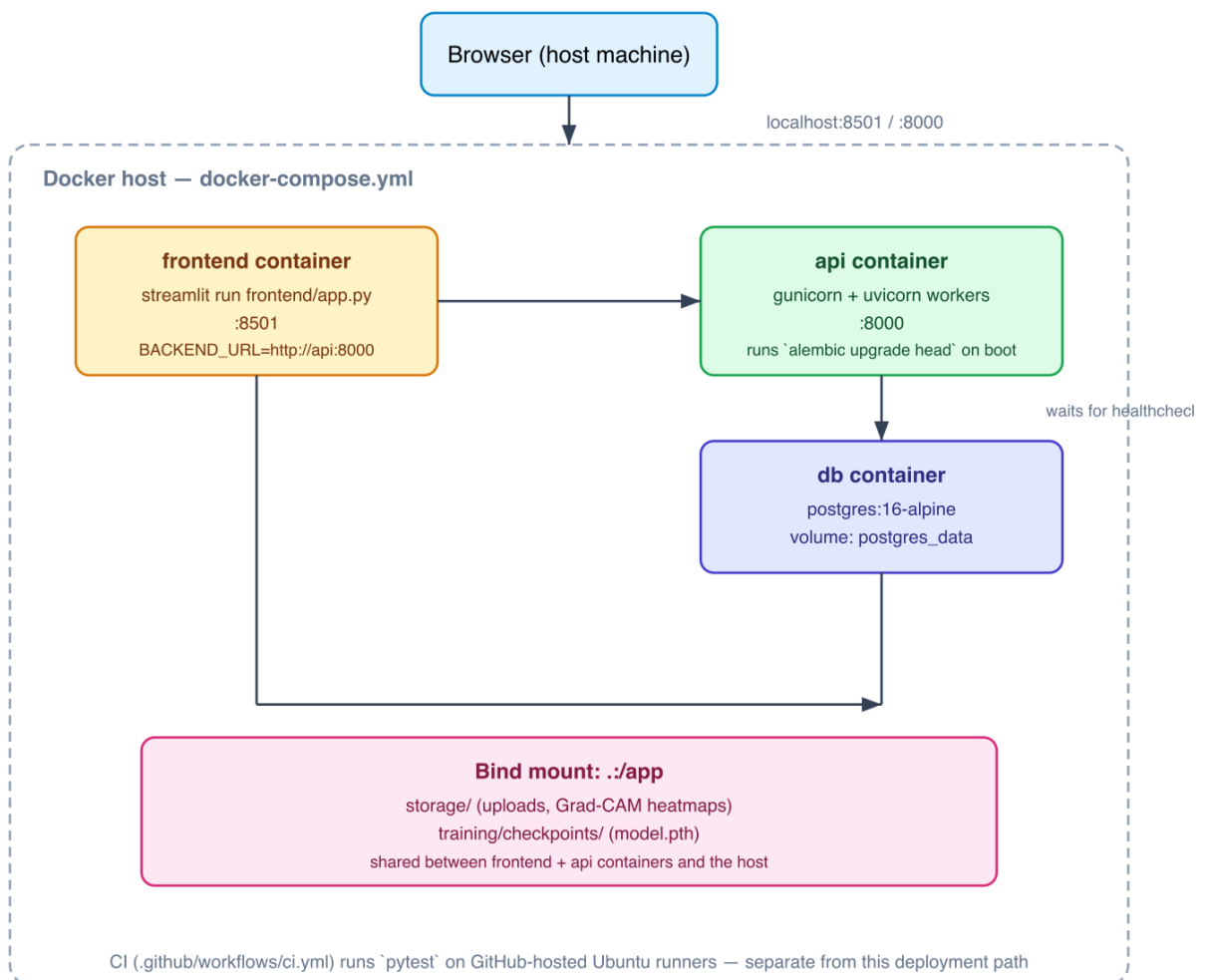
Grad-CAM runs on the trained model's final convolutional block (`layer4`) for the top-predicted class, producing a heatmap overlay saved alongside the prediction. This was generated and visually inspected against a real X-ray during development, not just unit-tested for output shape — see the worked example in model.md (model.md). It has not been reviewed by a radiologist for clinical plausibility; that gap is stated directly in known-limitations.md (known-limitations.md) rather than implied to be more than it is.

9. LLM Integration

`ReportService` sends the sorted disease-probability findings to the LLM configured in `config.yaml` (`gpt-4o-mini` by default) and returns a narrative report. The integration is designed around failure, not just success: if no `LLM_API_KEY` is configured or the call fails for any reason, the failure is caught and logged, `report_text` is `null`, and the rest of the prediction (probabilities + Grad-CAM) still returns successfully — a report should never be able to hide or block a diagnosis

that was already computed. This graceful-degradation path has been exercised for real (no API key is configured in the build/CI environment); the actual LLM call path is verified with a mocked client in tests, since no key was available to test against the real API — noted plainly rather than glossed over.

10. Deployment



Three Docker Compose services: **db** (Postgres 16, healthchecked), **api** (Gunicorn + Uvicorn workers, runs `alembic upgrade head` on boot), **frontend** (Streamlit, reaching **api** over the Docker network). Full detail, including two real bugs caught only by actually running the containers (a missing system library crash-looping the container, and an obsolete Compose config key), is in `deployment.md` (`deployment.md`).

11. Testing Strategy

30 automated tests (`pytest` , run on every push/PR to `main` / `develop` via GitHub Actions) cover model construction, preprocessing, the dataset loader, the training loop, every service

(`ImageService` , `PredictionService` , `GradCAMService` , `ReportService` , `StorageService`), every API route (with dependency-injected throwaway databases, not a shared test DB), Alembic migrations (upgrade *and* downgrade), and a Streamlit `AppTest` smoke test for the frontend.

Beyond automated tests, every feature in this project was **manually verified against a real running server** before being considered done — not just "tests pass." That includes: real `curl` calls against a live `uvicorn` process, a full build-and-run of the actual Docker Compose stack (including a genuine Postgres container, verified via direct `psql` queries, not just application logs), and a real trained checkpoint run through prediction and Grad-CAM on an actual image. Real bugs were caught this way that no unit test would have found — the `libGL.so.1` container crash-loop (Section 10) and a `KeyError: 'User'` from an unregistered SQLAlchemy model (`docs/database.md`) both surfaced only when the running system was actually exercised. A complementary, non-runtime bug — training augmentation silently never being applied, present since the first preprocessing implementation — was instead caught by a code review, and confirmed empirically before being trusted (see `model.md` (`model.md`)). Between the two, the lesson is the same: verify claims (whether "the tests pass" or "this looks right"), don't just make them.

12. Results

- Full pipeline (`upload` → `predict` → `Grad-CAM` → `report` → `history`) works end-to-end over real HTTP, against real Postgres, in real Docker containers — not just in isolated unit tests.
- A real (if small) training run on genuine chest X-ray data shows the model learning: best val AUC 0.71 (epoch 4 of 10, seeded run), with a reproducible overfitting pattern past that point.
- Grad-CAM produces a real, visually-inspected heatmap overlay for a real prediction.
- Report generation's failure path (no LLM configured) was exercised for real and confirmed not to break the rest of the response.
- 30/30 automated tests pass; CI is green on every push.

13. Known Limitations

Full list, with reasoning, in `known-limitations.md` (`known-limitations.md`). Summary:

- Only ~2,000 training images used, and the model overfits within ~5 epochs at that scale — proves the pipeline works, not a clinically meaningful model.
- No clinical validation of Grad-CAM output; not FDA approved; educational purpose only.
- **No authentication or access control on any endpoint** — the single biggest gap for anything beyond a local demo. `Prediction.user_id` and the `users` table exist in the schema but nothing populates or checks them yet.
- Audit logging limited to request-level middleware (method/path/status), not a structured per-inference "who viewed what" trail.
- No PACS/DICOM integration — accepts plain PNG/JPEG, not the format real radiology systems use.

- No model versioning — only one active checkpoint at a time (see `system-design.md` (`system-design.md`)'s scaling section for the concrete fix).

14. Future Improvements

- **Authentication (JWT/OAuth2)** on all patient-data endpoints — the clear top priority.
- **Scale up real training data further** (tens of thousands of images, stratified train/val split, explicit early stopping) toward the documented $AUC \geq 0.90$ target — 2,000 images improved on the original 300 but still overfits within ~5 epochs.
- **Model versioning** (`model_versions` table + `Prediction.model_version_id`) so multiple trained models can coexist and every historical prediction stays attributed to the model that produced it.
- **Structured audit logging** for compliance-grade traceability of who accessed which patient's data.
- **Object storage** (S3/GCS) instead of a local bind-mounted `storage/` directory, for durability and multi-host deployment.
- **DICOM/PACS support** to integrate with real radiology workflows instead of plain image uploads.
- **Async/queued inference** so a slow model call doesn't hold an HTTP worker for the duration of a prediction.

15. Conclusion

MedX AI demonstrates a complete, honestly-documented pipeline from image upload to explained, reported, persisted prediction — built the way a production system would be built (layered architecture, migrations, CI, containerized deployment) rather than as a single training notebook wrapped in an API. Its value is as much in the engineering discipline (small reviewed PRs, real-system verification catching real bugs, limitations stated rather than hidden) as in the ML itself. The foundation — API contracts, data model, service boundaries, deployment shape — is built to scale in the directions Section 14 describes, without needing to be re-architected first.